

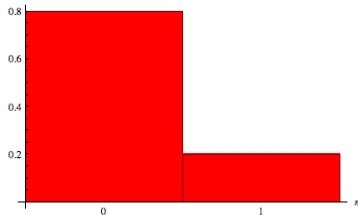
The Bernoulli Distribution

$$Pr(Y = y|p) = \begin{cases} 1 - p & \text{if } y = 0 \\ p & \text{if } y = 1 \end{cases}$$

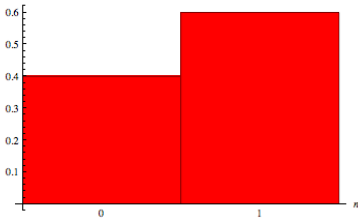
or

$$Pr(Y = y|p) = p^y(1 - p)^{1-y}$$

$P(n)$ for $p = 0.2$



$P(n)$ for $p = 0.6$



The Bernoulli distribution could describe the probability that a voter will vote or not vote.

The Binomial Distribution

$$Pr(Y = y|n, p) = \binom{n}{y} p^y (1 - p)^{n-y}$$

where n is the number of cases, y is the number of positive outcomes, and p is the probability of positive outcomes.

- This can provide the distribution of the number of heads that turn up for 20 coin tosses
- Each observation is composed of a binary outcome
- The observations are independent
- A record of the number of times one value was obtained

The Binomial Distribution Example

Consider three coin tosses, and say we are interested in the distribution of the number times we expect to see heads.

Toss 1	Toss 2	Toss 3	Number of Heads
T	T	T	0
H	T	T	1
T	H	T	1
T	T	H	1
H	H	T	2
T	H	H	2
H	T	H	2
H	H	H	3

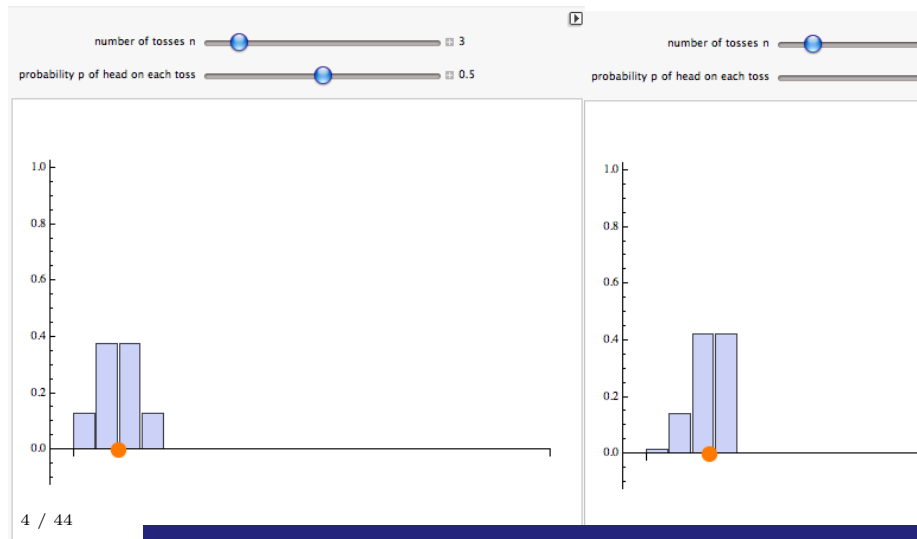
- What is the probability that we will observe 1 head?

$$Pr(Y = y|n, p) = \binom{n}{y} p^y (1 - p)^{n-y}$$

- $Pr(Y = 1|3, \frac{1}{2}) = 3(\frac{1}{2})(\frac{1}{2})^2 = \frac{3}{8}$

- $p(1 - p)(1 - p) + (1 - p)p(1 - p) + (1 - p)(1 - p)p = 3p(1 - p)^2$

Plotting the Binomial Distribution

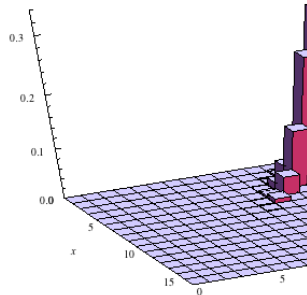
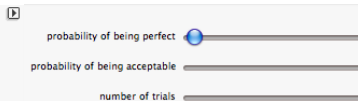
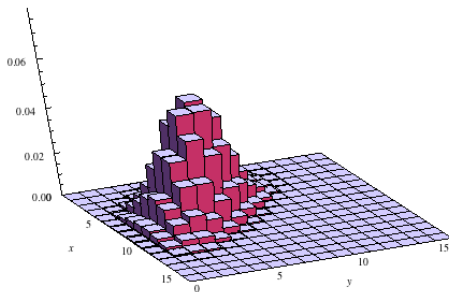


The Multinomial Distribution

The **multinomial distribution** counts the the number of times each of k mutually exclusive and collectively exhaustive outcomes happens, where each outcome happens with a probability of p_i , $i \in \{1, \dots, k\}$. Since the outcomes are mutually exclusive and collectively exhaustive $\sum_{i=1}^k p_i = 1$. Let Y_i represent the random variable that counts the times outcome i occurs. If there are n independent events then $Y_i \in \{0, 1, 2, \dots, n\}$ for all i and $\sum_{i=1}^k Y_i = n$. We write the PMF for non-negative integers $y_1, \dots, y_k$ as

$$Pr((Y_1 = y_1) \cap \dots \cap (Y_k = y_k)) = \begin{cases} \frac{n!}{y_1! \dots y_k!} \prod_{i=1}^k p_i^{y_i} & \text{if } \sum_{i=1}^k y_i = n \\ 0 & \text{if otherwise} \end{cases}$$

Plotting Trinomial Distribution



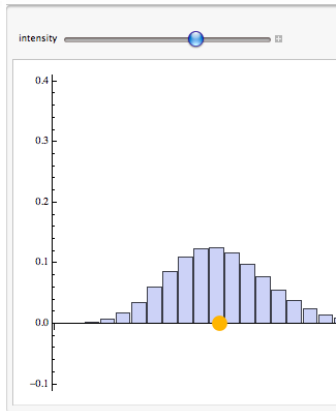
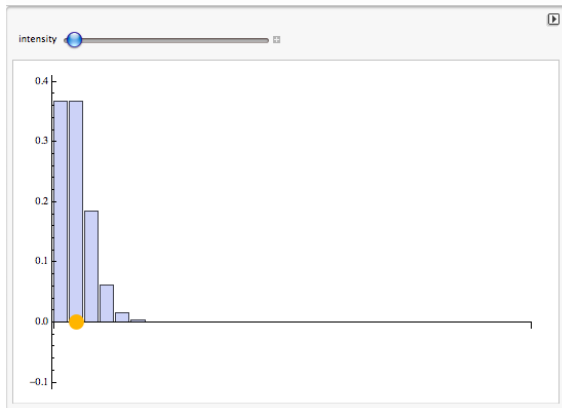
Event Count Distributions

The PMF of the Poisson Distribution can be written as:

$$Pr(Y = y|\mu) = \frac{\mu^y}{y! \times e^\mu}$$

Where μ is the expected number of events, y is a positive integer representing the number of events observed, and the variance σ^2 is equal to the mean, μ . This distribution provides the probability that we observe y events in a set amount of time given the expected number of events is μ .

Plotting Poisson Distribution



Assumptions of a Poisson Distribution

- Individual events must be countable as whole numbers over a given amount of time.
- The events must be produced independently from one another over a period of time.
- Third the average frequency of events must be known
- Could a Poisson Process produce the following counts?
 - Traffic Accidents
 - Civil War Deaths in a Year
 - Rain Storms
 - Primary victories for a US Presidential election.

Negative Binomial Distribution

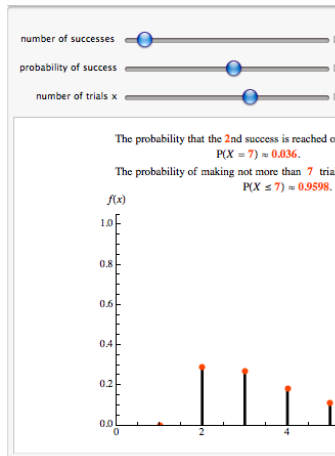
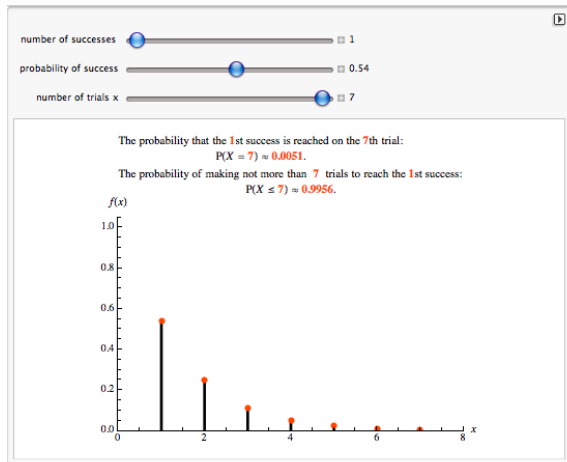
The PMF of the Negative Binomial Distribution can be written as

$$Pr(Y = y|r, p) = \binom{y + r - 1}{y} p^y (1 - p)^r$$

where y is the number of observed events (or successes), r is the number of observed non-events over $y + r$ opportunities, and p is the probability of any particular event (or success). The Negative Binomial provides a measure of the probability that we will observe y successes until a fixed number of failures are reached. For example,

- What is the probability of observing 1000 military casualties before one cease fire is brokered?
- What is the probability of flipping 5 heads before we flip two tails when flipping a coin?
- What is the probability that Brazil will play in 5 World Cups before winning the world cup?

Plotting Negative Binomial Distribution



Expectations of Random Variables

The **expectation** of a random variable X , denoted as $E_X[X]$ or simply $E[X]$ is the weighted average value that a random variable can take, where the weights are given by the probability distribution. Consider the expected value of a lottery, where a lottery consists of a set of values of outcomes and a corresponding set of probabilities.

- What is the expected value of a lottery that has the outcomes of \$0, \$100, and \$1000 at the corresponding probabilities of .989, .01, and .001?
- The expected outcome is $.989(\$0) + .01(\$100) + (.001)\$1000 = \2

Formula for Expected Value

$$E[X] = \sum_i x_i (Pr(X = x_i))$$

Computing Expected Number of Events

Say that a traffic czar wants to calculate the expected number of meter violations, if she changes the time limit on meters from 2 hours to 1 hour. We can calculate the expected number of meter violations for a time limit of 1 hour using the Poisson distribution.

$$E[X] = \sum_i i(Pr(X = i)) = \sum_{i=0}^{\infty} i \frac{\mu^i}{i! \times e^{\mu}} = e^{-\mu} \sum_{i=0}^{\infty} \frac{\mu^i i}{i!}$$

- $\sum_{i=0}^{\infty} \frac{\mu^i i}{i!} = 0 + \sum_{i=1}^{\infty} \frac{\mu^i i}{i!}$ (Calculating the first term in the series)
- $= \sum_{i=1}^{\infty} \mu \frac{\mu^{i-1}}{(i-1)!}$ (Factoring out μ and dividing through by i)
- $= \mu(\sum_{i=0}^{\infty} \frac{\mu^i}{i!})$ (Recognizing an infinite sum from 1 to ∞ of $i - 1$ is equivalent to the infinite sum of from 0 to ∞ of i)
- $= \mu e^{\mu}$ (Using the Taylor series definition of e)
- $E[X] = e^{-\mu} \mu e^{\mu} = \mu$

Expected Utility

$$EU(X) = \sum_i u(x_i)(Pr(X = x_i))$$

where $u(x_i)$ is the Bernoulli utility function and represents the know utility from the value that a random variable can take.

$EU(X)$ is the Von Neumann-Morgenstern utility function and represents the expected utility that one derives from a lottery.

Paper, Rock, Scissors Example

Consider two people, Lisa and Tom, who are playing Paper, Rock, Scissors. The loser of the game must pay the winner \$1. If there is a tie, no payments are distributed. Say that Tom plays each action with a probability of $\frac{1}{3}$, and that Lisa's utility function is $u_l(x) = x$ where x can take on the value of -1, 0, or 1. What is Lisa's expected utility from playing Rock?

$$\blacksquare U(R, \sigma_T) = \frac{1}{3}u_l(R, S) + \frac{1}{3}u_l(R, R) + \frac{1}{3}u_l(R, P) = \frac{1}{3}(1 + 0 - 1) = 0$$

Preferences over Risk

Consider the lottery that pays \$0 at a probability of $\frac{1}{2}$ and pays \$10 at a probability of $\frac{1}{2}$.

- A person is **Risk Averse** if she prefers to take \$5 over the lottery. This person would have a *concave* utility function.
- A person is **Risk Neutral** if she is indifferent between the lottery and taking \$5. This person would have a linear utility function.
- A person is **Risk Seeking** if she prefers the lottery over taking \$5. This person would have a convex utility function

Preferences over Risk

Consider the lottery that pays \$0 at a probability of $\frac{1}{2}$ and pays \$10 at a probability of $\frac{1}{2}$.

- $u(x) = x^{\frac{1}{2}} : EU(X) = \frac{1}{2}(0)^{\frac{1}{2}} + \frac{1}{2}(10)^{\frac{1}{2}} < 5^{\frac{1}{2}}$
- $u(x) = x : EU(X) = \frac{1}{2}(0) + \frac{1}{2}(10) = 5$
- $u(x) = x^2 : EU(X) = \frac{1}{2}(0)^2 + \frac{1}{2}(10)^2 > 5^2$

Moments of a Distribution

The moments of a distribution can be used to describe a distribution. The k th moment of the random variable X can be represented as $E[X^k]$, where $E[\]$ represents the expectation of the function in the brackets. Thus, the mean, $E[X]$ is the first moment of the variable. The moment about zero is defined as

$$\sum_i x_i^k (Pr(X = x_i))$$

Note that if $k = 1$ then the first moment about zero just provides the mean or the central tendency of the distribution

Variance

We can also calculate the moments about the mean of X .

$$\sum_i (x_i - \mu)^k (Pr(X = x_i))$$

The second moment about the mean provides the variance of a distribution, which measures the spread of the distribution.

Skewness

The third moment about the mean, also denoted as μ_3 and divided by the standard deviation (the square root of the variance) defines the skewness:

$$E[(\frac{X-\mu}{\sigma})^3] = \frac{\mu_3}{\sigma^3}$$

If skewness is zero the the distribution is perfectly symmetric around the mean.

Kurtosis

The fourth moment about the mean and divided by the standard deviation defines the kurtosis. The kurtosis provides a measure of the flatness or peakedness of the distribution.

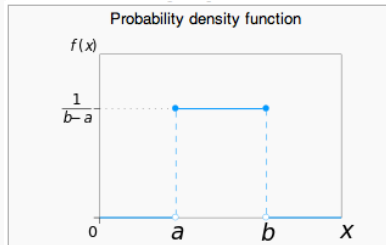
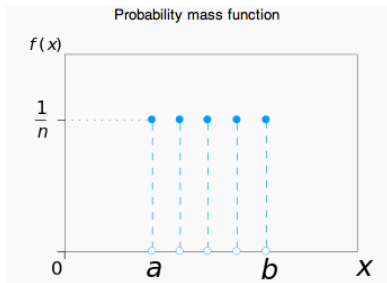
$$E[(\frac{X-\mu}{\sigma})^4] = \frac{\mu_4}{\sigma^4}$$

Probability Distribution Function

A **Probability Distribution Function** is a function that describes the curve that connects the various probabilities of specific ranges of values that a random variable can assume.

- The PDF differs from the from the PMF in that it does not describe the chance that any particular value of the random variable is drawn at random from the distribution. Rather, it describes the relative likelihood of drawing any specific variable, and the exact probability of drawing a value within some range.

Difference between PMF and PDF of Uniform



In the graph on the right the probability that $x \in [a, b] = 1$. The probability that $x \in [a, a + \epsilon]$, where $\epsilon > 0$ and $a + \epsilon < b$ is greater than zero but less than one. The probability that $x = a$ is not well defined.

PDF for a Continuous Variable

For a continuous variable,

$$P(X \in [a, b]) = \int_a^b f(x) dx$$

where $f(x)$ is the PDF for the random variable X . The probability of drawing an x in the interval from a to b is equivalent to the area underneath the curve of the PDF from a to b .

- What is $P(-\infty < X < \infty) = \int_{-\infty}^{\infty} f(x) dx$?

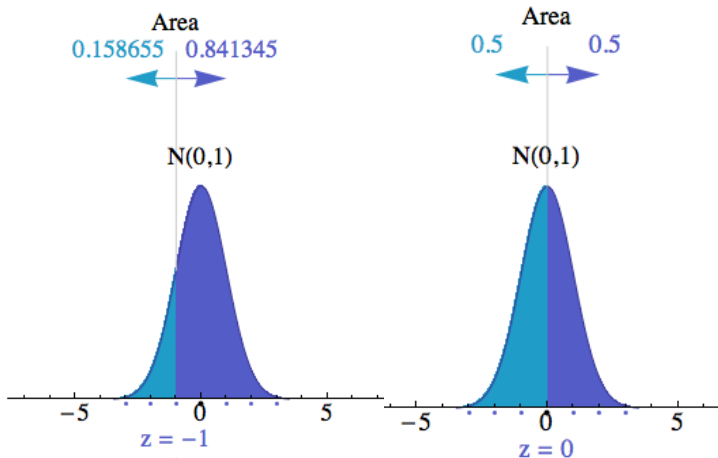
CDF for a Continuous Variable

The CDF measures the probability that $X \leq x$. So we write the CDF as:

$$P(X \leq x) = F(x) = \int_{-\infty}^x f(t) dt$$

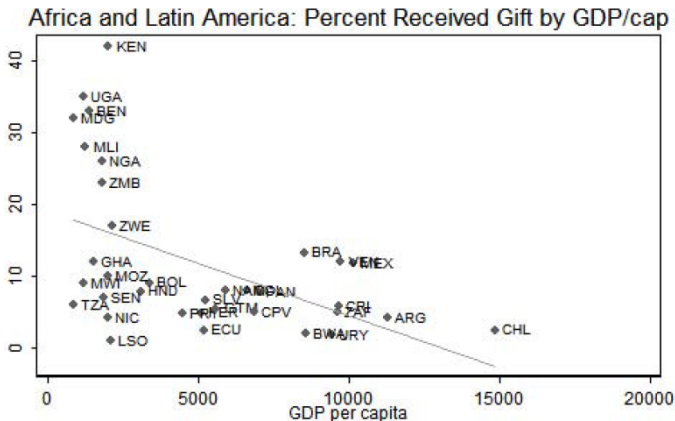
The CDF is the anti-derivative of the PDF. It must take on values within the interval of 0 to 1.

Visualizing the CDF and PDF for the Standard Normal



Joint Distribution Scatterplot

When we want to examine the joint distribution of continuous variables, or continuous and discrete variables, we use a scatterplot.



Theoretical Joint Distributions

For two discrete random variables we can write their joint PMF as $f(x, y) = Pr(X = x \cap Y = y)$, which is the chance that both x and y are simultaneously realized. For example the joint probability of rolling two 1s with two dice is $\frac{1}{36}$. For a joint probability of a continuous variable we can write:

$$Pr(X \in [a, b] \cap Y \in [c, d]) = \int_c^d \int_a^b f(x, y) dx dy$$

We can write the CDF as:

$$Pr(X \leq a \cap Y \leq b) = \int_{-\infty}^b \int_{-\infty}^a f(x, y) dx dy$$

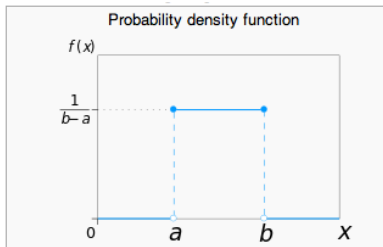
Theoretical Joint Distributions Continued

- If the random variables X and Y are independent then
$$f(x, y) = f(x)f(y)$$
- If X is conditional on Y then $f(x, y) = f_{X|Y}(x|y)f_Y(y)$
- $f_Y(y)$ is the marginal distribution of the random variable Y . This means that $f_Y(y) = \int f(x, y)dx$
 - Here thinking in the analogy to discrete terms is helpful. Consider calculating the joint probability of rolling a die and drawing a number from 1 to 4. If we want to calculate the marginal probability of rolling a 1, we would sum up the joint probability of rolling 1 and drawing a 1, rolling a 1 and drawing a 2, rolling a 1 and drawing a 3, and rolling a 1 and drawing a 4. This is just $\frac{4}{24} = \frac{1}{6}$
- The conditional distribution is $f_{X|Y}(x|y) = \frac{f(x,y)}{f_Y(y)}$

Expectations of Continuous Random Variables

- Recall that we can write the expectation of a discrete random variable as $E_X[X] = \sum_i x_i Pr(X = x_i)$ or if the PMF is $f(x_i)$, we have $E_X[X] = \sum_i x_i f(x_i)$.
- For a continuous random variable the PDF is
$$E_x[X] = \int_{-\infty}^{\infty} xf(x)dx$$
- Expected utility for a continuous random variable is
$$E_x[X] = \int_{-\infty}^{\infty} u(x)f(x)dx$$

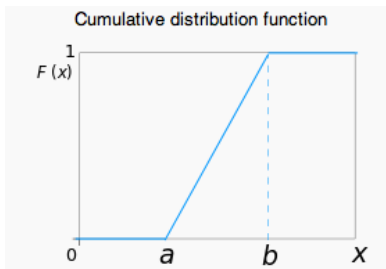
The Uniform Distribution



$$f(x) = \begin{cases} \frac{1}{\beta - \alpha} & \text{if } x \in [\alpha, \beta] \\ 0 & \text{if otherwise} \end{cases}$$

- It is the case that $\int_{\alpha}^{\beta} \frac{1}{\beta - \alpha} dx = \frac{x}{\beta - \alpha} \Big|_{\alpha}^{\beta} = \frac{\beta - \alpha}{\beta - \alpha} = 1$
- α is the location parameter and β is the scale parameter.
- When X is distributed uniformly with these parameters we write $X \sim U[\alpha, \beta]$

The CDF of the Uniform Distribution



The CDF of the uniform distribution is defined as:

$$F(x) = \begin{cases} 0 & \text{if } x < \alpha \\ \frac{x-\alpha}{\beta-\alpha} & \text{if } x \in [\alpha, \beta] \\ 1 & \text{if } x > \beta \end{cases}$$

Moments of Continuous Distributions

The k th moment about zero is

$$\int_{-\infty}^{\infty} x^k f(x) dx$$

The k th moment about the mean is

$$\int_{-\infty}^{\infty} (x - \mu)^k f(x) dx$$

- What is the mean of a distribution?
- What is the variance of a distribution?

Covariance

Covariance consider the variation around the mean for two variables.

$$\sigma_{xy} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x - \mu_x)(y - \mu_y)f(x, y) dx dy$$

Where μ_x is the mean of X and μ_y is the mean of Y .

- What is the value of the integrand when both variables exceed the mean or are below the mean?
- What is the value of the integrand when one variable exceeds the mean and the other variable is below the mean

Stochastic Dominance

- Say $f(x)$ and $g(x)$ are two different PDFs. $f(x)$ first-order stochastically dominates FOSD $g(x)$ if for the CDFs we have $F(x) \leq G(x)$. Note it must also be the case that $1 - F(x) \geq 1 - G(x)$.
 - Consider lottery 1 where the chances of winning \$5 is .25, \$10 is .5, and \$100 is .25. Consider lottery 2 where the chances of winning \$0 is .25, \$5 is .25, and \$10 is .5. Does one lottery FOSD the other lottery?
- It is also the case that if $f(x)$ FOSD $g(x)$ then $\int_{-\infty}^{\infty} u(x)f(x)dx \geq \int_{-\infty}^{\infty} u(x)g(x)dx$ for all increasing functions $u(x)$.

See pg. 258 for Second Order Stochastic Dominance.

The Normal Distribution

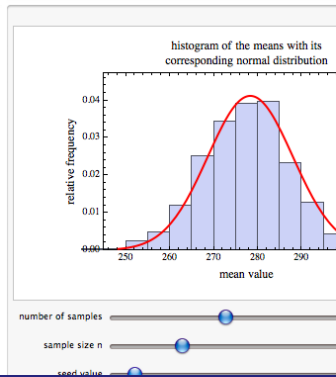
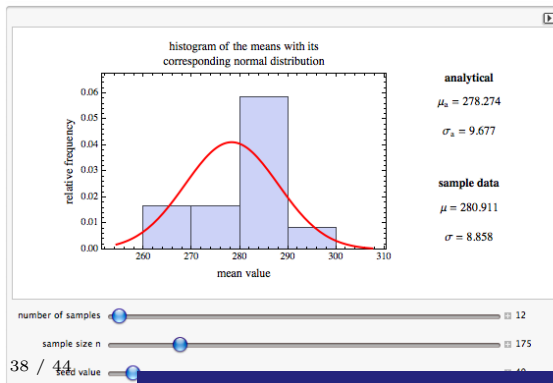
If X is distributed normally then we write $X \sim N(\mu, \sigma^2)$, where the mean of this distribution is μ and the variance around the mean is σ^2 . The PDF of this distribution is

$$f(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$$

- μ and σ^2 are independent.
- The distribution is symmetric around the mean.
- What are some limitations of this distribution for describing social processes?
- Error terms in regression analysis are assumed to be distributed normally.

Central Limit Theorem

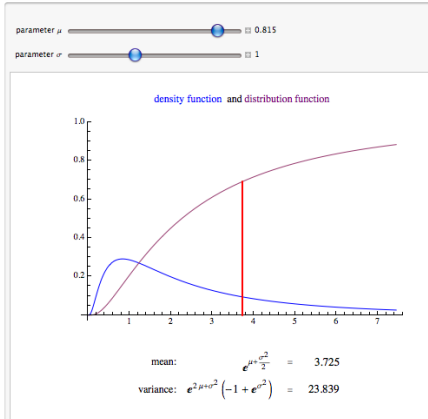
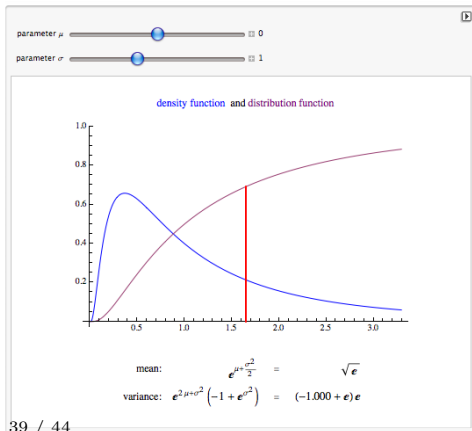
The central limit theorem states that the distribution of the sum (or average) of a large number of independent, identically distributed variables will be approximately normal, regardless of the underlying distribution.



Log-Normal

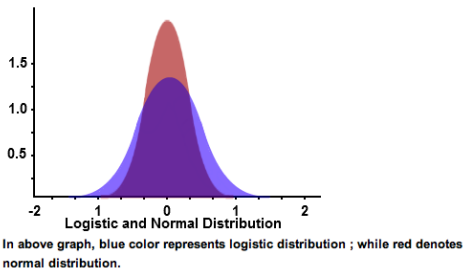
The PDF of the Log-Normal distribution is

$$f(x; \mu, \sigma^2) = \frac{1}{x\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(\ln(x)-\mu)^2}$$



Logistic Distribution

The logistic distribution is also defined by its mean and variance, and looks similar to the normal, but it has larger tails.



Duration Distributions

- Duration Distributions can be used to measure concepts that are distributed over time
 - For example, Regime Survival, Civil War Duration, and Parliamentary Coalition Survival.

The Exponential Distribution

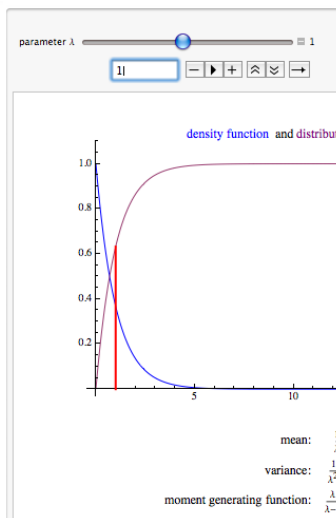
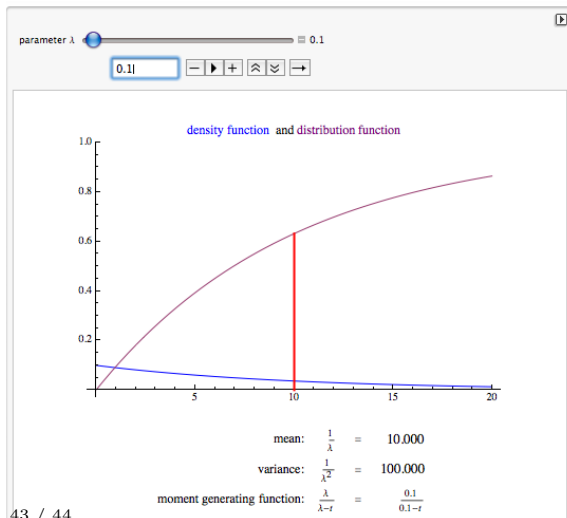
The PDF for the exponential distribution is

$$f(x; \mu) = \frac{1}{\mu} e^{-\frac{x}{\mu}}$$

where μ is the mean duration of events.

- The Exponential Distribution function measures the distribution of the duration of an event
- Requires that the risk of failure is constant over time.
 - What is problematic about this assumption for political science?

Plotting Exponential Distribution



The Pareto Distribution

The PDF for the Pareto Distribution can be expressed as

$$f(x; \kappa, \beta) = \begin{cases} \frac{\kappa \beta^\kappa}{x^{\kappa+1}} & \text{for } x \geq \beta \\ 0 & \text{for } x < \beta \end{cases}$$

where $\kappa > 0$ is a shape parameter and β is a scale parameter. This distribution assumes that initial values have an effect on the size of subsequent values.